

1. A particle of mass 6 kg moves such that $\mathbf{r} = \begin{pmatrix} t^2 - 5 \\ t^2 - 3t + 2 \end{pmatrix}$.
 - (a) Calculate the times when the particle crosses the x -axis.
 - (b) Find an expression for $\mathbf{v}$ in terms of t , and hence calculate the speed of the particle at $t = 6$ s.
 - (c) Find $\mathbf{a}$ and hence calculate the constant force acting on the particle.

2. A particle of mass 4 kg moves such that $\mathbf{v} = \begin{pmatrix} 3t^2 - 2t - 4 \\ 3t^2 - 4t + 3 \end{pmatrix}$. Initially the particle is at the point $(3, -7)$.
 - (a) Find an expression for $\mathbf{r}$, the displacement of the particle, after a time t .
 - (b) Calculate the times when the particle crosses the line $y = x$.
 - (c) Find an expression for $\mathbf{a}$ in terms of t and hence calculate the force acting on the particle at $t = \frac{2}{3}$ s.

3. A particle P moves such that at time t seconds its position vector, $\mathbf{r}$ metres, is given by:

$$\mathbf{r} = \begin{pmatrix} t^3 - 4 \\ t^2 - t \\ t^2 - 8t + 5 \end{pmatrix}$$
 - (a) By first finding an expression for $\mathbf{v}$, the velocity of the particle, in terms of t calculate the speed of P when $t = 2$.
 - (b) Find the value of t for which the z -coordinate of $\mathbf{r}$ is a minimum and find the position of P at this time.
 - (c) Find the average velocity of P from $t = 1$ to $t = 3$.

4. A particle moves such that $\mathbf{v} = \begin{pmatrix} 3t^2 \\ 4 - 2t \\ 4t^3 \end{pmatrix} \text{ ms}^{-1}$. Initially the particle is at the point $(0, 0, -1)$. Find:
 - (a) the acceleration when $t = 2$ s.
 - (b) the distance of the particle from the origin when $t = 3$
 - (c) the maximum value of the y -coordinate of the position of the particle
 - (d) the average velocity between $t = 1$ to $t = 3$ s.

5. A particle moves such that its position can be well modelled by $\begin{pmatrix} t^2 + 5t - 1 \\ t^3 + 4t^2 - 9t + 11 \end{pmatrix}$.
 - a. Find the time when the particle is travelling on a bearing of 45° .
 - b. Find the times (if any) when the particle is travelling directly towards the origin.
 - c. Find the maximum distance of the particle from the origin in the interval $0 \leq t \leq 10$.
 - d. Find the times (if any) when the particle appears to be getting no further from the origin.